CSA04-Operating Systems

TASK 11 – Recommend and Defend the Operating-System Plan

1. Recommended Operating-System Plan for the Hospital

The hospital requires a reliable, secure and efficient operating-system plan to manage patient monitors, MRI scanners, database servers and hospital applications.

I recommend using different operating-system environments based on the requirements of each device.

2. Type and Structure of Operating System

Environment	Recommended OS	Structure	Reason
Patient Monitors	Real-Time Operating System (RTOS)	Microkernel	Provides fast and predictable responses.
MRI Scanner	Embedded Linux	Modular	Supports device drivers and high-speed data transfer.
Database Server	Linux Server	Monolithic Modular	Provides stability, performance and security.
Hospital Application Server	Linux Server	Modular	Supports multiple applications and users.
Backup Server	Linux Server	Modular	Provides reliable storage and backup management.

3. Services, Protection and I/O Methods

Environment	OS Services	Protection	I/O Method
Patient Monitors	Scheduling and interrupt handling	User/kernel mode and memory protection	Interrupt-driven I/O
MRI Scanner	File management and device management	Access control and memory protection	DMA
Database Server	Process, memory and file management	Authentication and permissions	DMA and interrupt-driven I/O
Application Server	Networking and process management	Firewall and access control	Interrupt-driven I/O
Backup Server	Storage and file management	Encryption and access control	DMA

Explanation:

- Patient monitors require interrupt-based processing for quick emergency responses.
- MRI scanners require DMA to transfer large medical images efficiently.
- Database servers require caching to improve patient record retrieval.
- Application servers require process scheduling and network services.

- Backup servers require reliable storage and data protection.

4. Limitations and Risks

1. Hardware failure: Failure of a monitor or server may interrupt hospital operations.
2. Cybersecurity threats: Unauthorized access may expose confidential patient records.
3. Network failure: Communication between hospitals may be interrupted.
4. Virtualisation overhead: Virtual machines may consume additional CPU and memory resources.
5. Maintenance cost: Different operating systems require skilled administrators and regular updates.

These risks can be reduced through regular backups, security updates, redundant hardware and continuous monitoring.

5. Suitability for Three Hospitals

The proposed plan is suitable for a group of three hospitals because:

- Each hospital can operate its local monitoring systems independently.
- Patient records can be shared through a centralized database.
- Network storage supports data sharing between hospitals.
- Centralized backup improves disaster recovery.
- Standardized operating systems simplify maintenance and administration.

Therefore, the plan provides scalability, reliability and efficient resource management across all three hospitals.

6. Alternative Concept and Complexity Analysis – Virtual Machines

If the complete hospital system is moved to virtual machines running on two large physical servers, a hypervisor can manage multiple virtual machines.

Recommended approach:

- Server 1: Database and application virtual machines.
- Server 2: Backup and secondary application virtual machines.
- Patient monitors: Prefer dedicated physical hardware or validated real-time environments rather than relying on ordinary virtual machines.
- MRI scanners: Use dedicated hardware interfaces or carefully validated device passthrough.

Virtualisation Architecture

Hospital Applications

Patient records • MRI • Monitoring • Backup

Physical Server 1

Virtual Machine 1

Database

Virtual Machine 2

Applications

Physical Server 2

Virtual Machine 3

Backup

Virtual Machine 4

Secondary Apps

Hypervisor Layer

Allocates CPU, memory and hardware resources

Physical Hardware

Two servers • CPU • RAM • SSD • Network

Evaluation of Virtualisation

A. Response Time

- Virtualisation introduces some scheduling and device-access overhead.
- Patient monitors may experience unpredictable delays under heavy server load.
- Dedicated real-time hardware is safer for critical monitoring.
- CPU reservation and real-time scheduling can reduce delays but must be validated.

B. Memory and Processor Overhead

- Each VM requires memory for its guest OS and applications.

- Hypervisors consume additional CPU and memory resources.
- Multiple VMs share physical processors and storage.
- Resource overcommitment can reduce performance.

C. Reliability

- Virtual machines allow easier backup, migration and recovery.
- Two servers provide some redundancy.
- However, failure of one physical server may affect several VMs simultaneously.
- Replication, failover, independent backups and redundant networking are required.

Comparison

Factor	**Separate Physical Machines**	**Two Virtual Servers**
Response time	More predictable	Depends on VM load
Memory usage	Separate resources	Shared physical RAM
CPU usage	Dedicated resources	Shared CPU resources
Maintenance	More difficult	Easier management
Reliability	Better isolation	Risk of host failure
Cost	Higher hardware cost	Lower hardware cost

Evaluation: Virtualisation is suitable for database, application and backup servers because it improves resource utilization and management. However, critical patient monitors should preferably remain on dedicated real-time systems to ensure predictable response times.

7. Final Recommendation and Conclusion

1. Use RTOS for patient monitoring to ensure fast and predictable responses.
2. Use embedded Linux for MRI scanners and DMA for large image transfers.
3. Use modular Linux servers for databases, applications and backups.
4. Apply memory protection, authentication, caching and regular backups.
5. Use virtualisation for non-critical server workloads while keeping critical monitoring systems isolated.

Final Result: The recommended operating-system plan provides security, reliability, performance and scalability for all three hospitals. The analysis of system calls, memory hierarchy, interrupts, process creation and I/O performance supports selecting different operating environments according to each hospital device's requirements.